HARDWARE & MIXED-SIGNAL ENGINEERING PORTFOLIO SUMMARY

Design & Construction of an 8-Step Analog Tone Sequencer

B.Eng. Final-Year Project • Department of Computer Engineering, University of Benin (UNIBEN)

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TOPOLOGY	ACTIVE ICS	CHANNEL ISOLATION	SUPPLY & DECOUPLING	IMPLEMENTATION
Mixed-Signal Analog/CMOS	2× NE555D + 1× CD4022B	8× 1N4148 Diode Steering	9 V DC | 0.1 µF & 10 µF Bypass	Sim → Breadboard → Veroboard

1. MIXED-SIGNAL ARCHITECTURE & TIMING FORMULATION

Bipolar Timer / CMOS Johnson Counter Interface

This project demonstrates the end-to-end design, simulation, physical prototyping, and characterization of a discrete 8-step mixed-signal tone sequencer. Rather than relying on software-abstracted microcontroller PWM, the architecture directly interfaces continuous-time analog relaxation oscillators with synchronous CMOS sequential logic. An astable master clock (`U1: NE555D`) drives an octal Johnson counter (`U3: CD4022B`), whose decoded one-hot outputs (`Q0–Q7`) sequentially bias an 8-channel `1N4148` diode-steered resistive matrix (`RV3–RV10`). Each active channel modulates the charging current of a second astable tone oscillator (`U2: NE555D`) to synthesize distinct, user-programmable audio frequencies across eight time-multiplexed steps.

$$f_{clk} = 1.44 / [(R_1 + 2R_{eff}) \times C_1]$$ (Master Clock)

$$f_n = 1.44 / [(R_A + 2R_n) \times C_5]$$ (Step-n Tone, n ∈ {0…7})

$$D = [(R_A + R_n) / (R_A + 2R_n)] \times 100\%$$ (Duty Cycle)

2. HARDWARE CIRCUIT SCHEMATIC & SIGNAL CHAIN

Reconstructed from Verified Prototype (Figure 4.1)

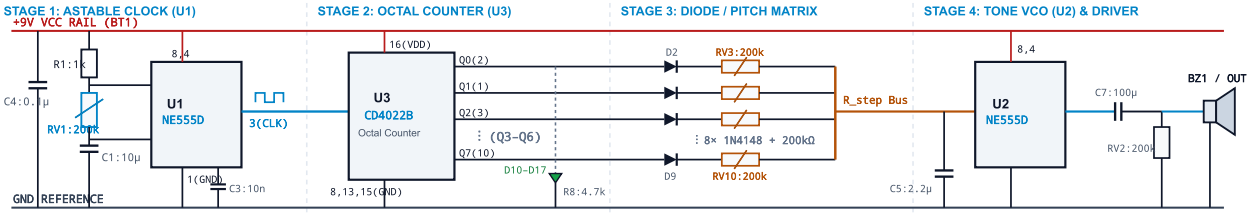


Figure 1: Complete hardware schematic showing astable clock (`U1`), CMOS counter (`U3`), `1N4148` steering matrix, and VCO/RCO (`U2`).

Supply: 9 V DC Regulated/Battery

3. SUBSYSTEM ENGINEERING, VERIFICATION FLOW & PRACTICAL SKILLS

End-to-End Hardware Bring-Up & Characterization

- **Mixed-Signal Domain Crossing & Diode Steering Isolation:** Interfaced the continuous-time bipolar clock output (`U1 Pin 3`) with the CMOS clock input (`U3 Pin 14`) using defined pull-down termination (`R3 = 47 kΩ`) to eliminate false edge triggering. Designed an 8-branch `1N4148` fast-switching diode steering array (`D2–D9`) to isolate inactive CMOS outputs and prevent reverse-current crosstalk across the `200 kΩ` pitch trimmers (`RV3–RV10`).
- **Power Integrity, Decoupling & Parasitic Mitigation:** Mitigated `NE555` comparator switching transients and supply-rail ripple by placing local `0.1 µF` ceramic (`C4, C8`), `0.01 µF` control-voltage (`C3`), and `10 µF` electrolytic (`C1, C2, C6`) bypass capacitors across IC supply pins. Transitioning from solderless breadboard to permanent copper stripboard (Veroboard) reduced stray inter-nodal capacitance and eliminated intermittent contact resistance.